Part E - Clinical figure justification

Panel 1 asks whether systolic pressure has one population or multiple modes. A 5 mmHg histogram exposes two modes; aligned position and bar length encode quantitative counts on a common zero baseline. I compared 3, 45 and 400 bins first. Bin edges 90-205 cover every value (n=10,000); the two-mode shape is not evidence identifying a specific clinical subgroup.

Panel 2 asks which wards have longer and more variable stays. A box plot compares median, IQR and tails. Ward order is ascending median; all four use the same blue because labelled positions identify category. Log10 y preserves the 273.65-day maximum while separating typical stays (overall median 1.38 days; skewness 22.755). Whiskers=1.5 IQR, width=.6, linewidth=1, fliersize=2; showfliers=True retains unusual observations.

Panel 3 asks how the strongest independently recorded pair behaves. Panel 4 screens all nine measurements: systolic/diastolic is strongest after excluding derived MAP. Scatter uses aligned x/y position, not area, to show association; it is not a novel causal discovery. A uniform sample of 2,000 valid pairs, seed=220, size=10 points squared, alpha=.18 and no point edges controls overlap. Pearson r uses all valid pairs. Age/cholesterol is a weaker additional association.

Panel 4 asks which independently recorded indicators correlate. Pearson correlation uses pairwise-complete data (minimum 3 pairs); sample-size matrix is exported. Lower triangle excludes diagonal and duplication; RdBu_r, vmin=-1, vmax=1 and center=0 preserve sign and comparability. Two-decimal annotations provide exact readings; cell linewidth=.3 and colourbar shrink=.8 improve legibility. MAP is arithmetic, and risk_score is a ratio, so neither enters the matrix.

Cleaning and exclusions: all 10,000 patient rows remain. Normalize 12 ward spellings to 4 (6,533 labels changed); parse three explicit date formats, including day-first slashes (10,000 successful; 0 failures). BMI 0 (84) and 999 (402) are missing sentinels: 486 cells set missing; mean 66.17704 to 27.34627 kg/m2. Heart-rate readings above 250 bpm were excluded as device faults (n=29), plus 8 zero readings invalidated. Mean HR moves from 78.83310 to 78.04687 bpm (n=9,963). The figure was also produced without the HR/BMI exclusions; it and the shared-scale HR sensitivity histogram are in the appendix.

Two zero-day LOS rows are omitted only from the log box plot (10,000 to 9,998; mean 2.309091 to 2.309553 days). Possible same-day stays and every long stay remain in the source and other analyses; no arbitrary upper LOS cutoff is imposed. Wide LOS is melted: 6,072 padding cells, not patient records, are omitted; 10,000 observations match ID-linked ward medians/counts. No other rows are deleted. The Scott SpO2 KDE leaks beyond 100%, so bounded empirical probabilities are used in Part A; vitals normality is not assumed.

Risk-score mean 1.120877 is not an aggregate event rate. Weighting by LOS gives only an approximation (.654373 events/day) because the exact event numerator is absent and the supplied ratio is rounded; no exact rate is claimed. These choices precede outcome interpretation and would remain if they weakened the finding (symmetry test). Figure: 15 x 11 inches, font DejaVu Sans, message titles, labelled units, top/right spines removed, tight layout, white background, 300 dpi and tight bounding box; source and sample rules accompany the export.

Source: supplied DATA_DICTIONARY.md, TASKS.md and vizlib.py; synthetic CDB601220 Week 3 data. No patient-risk prediction or real-world policy conclusion is claimed.